Temporal Partial Information Decomposition of Human EEG Across Sleep Stages

Short-window pass (1-second resolution)

Temporal Information Decomposition Project · June 2026
Subject set: Bitbrain ds005555 sub-1..sub-10 · 6 PSG channels · 3.5 h

Abstract

We re-apply *Temporal Partial Information Decomposition* (Temporal PID) to overnight polysomnography EEG from 10 healthy adults using a **1-second window** configuration that brings the analysis into a temporal scale appropriate for EEG dynamics. For every 1-second EEG window we construct temporal triplets at pairs of second-scale lags (τ_1, $\tau_2 \in$ [1, 30] s) and decompose the predictive information about the target window into four atoms — redundancy, synergy, unique information from the shorter lag (Unique_1), and unique information from the longer lag (Unique_2) — using the Minimum Mutual Information (MMI) PID measure, implemented in closed form (verified to 10^{-15} against the dit reference). The first 3.5 hours of each recording are analysed — enough for two NREM-REM cycles plus per-stage AR(1) fits.

This pass updates the 30-second report in three substantive ways. First, the block-permutation null now shuffles 1-second blocks, destroying nearly all linear autocorrelation at the tested lag scales rather than preserving it. Second, the AR(1) baseline is **stage-conditional**: ϕ and σ are fit per sleep stage on that stage's windows alone, isolating non-linear excess from across-stage spectral differences. Third, a new figure (the per-stage PID excess + double-dissociation strip) directly addresses the question of whether stages differ in *how* they exceed their own linear baseline, not just by how much.

Introduction

Sleep staging traditionally rests on spectral features such as slow-wave power, sleep spindle density, and K-complex count (Berry et al., 2012). These measures capture *linear, single-timescale* properties of the EEG. The rich temporal dynamics of sleep — travelling slow oscillations that coordinate memory consolidation, hippocampal sharp-wave ripples, and thalamo-cortical spindles — involve *multi-timescale, non-linear* interactions that spectral power alone cannot resolve.

Partial Information Decomposition (PID; Williams & Beer, 2010) is an information-theoretic framework that decomposes the total predictive information shared between a set of source variables and a target into non-negative, non-overlapping atoms. When applied across temporal lags (comparing a signal to its own past at two different delays), PID quantifies the temporal self-predictability structure of neural dynamics.

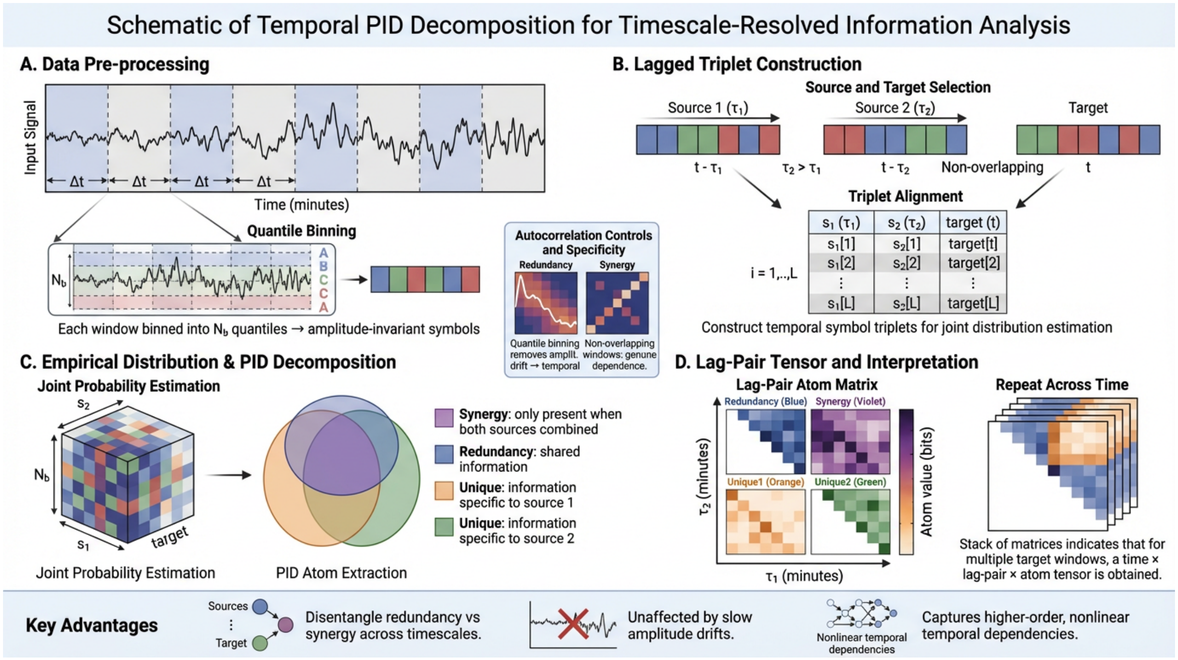
The companion 30-s pass established that all four PID atoms vary highly significantly across sleep stages on this dataset. Two methodological concerns motivated the present short-window pass. First, a 30-s block is long for EEG — the EEG meaningfully changes within a 30-s window, and a 30-s block-permutation null preserves nearly all the temporal structure of interest, weakening the null's informativeness. Second, the global (non-stage-conditional) AR(1) baseline used previously confounds within-stage non-linear excess with across-stage spectral differences. Both are addressed here.

Methods

Dataset and participants

EEG data were taken from OpenNeuro `ds005555` (10 healthy adults, overnight polysomnography). The first 3.5 hours of each recording were analysed — a reduction from 5 hours that still captures two full NREM–REM cycles while bounding wall-time at the 1-second resolution. Six scalp channels (F3, F4, C3, C4, O1, O2) were loaded with MNE-Python (Gramfort et al., 2013); sleep stages were read from the accompanying `*psg_events.tsv` files (AASM five-class system).

Figure 1: Overview of the Temporal PID pipeline.



(A) Input signal segmented into 1-s windows, independently discretised into N_b = 4 quantile bins. (B) Lagged triplet construction. (C) Empirical joint over N_b^3 = 64 symbol states; PID-MMI in closed form. (D) Tensor over time × lag pair × atom.

Preprocessing

Each EEG channel was bandpass filtered (0.5–60 Hz, 4th-order Butterworth, zero-phase) and notch filtered at 50 Hz and 60 Hz (IIR notch, Q = 30).

Windowing and Discretisation

The continuous signal was divided into non-overlapping windows of Δt = 1 s (reduced from 30 s in the original pass), yielding 12,600 windows over 3.5 hours. Each window was independently discretised into N_b = 4 levels using quantile binning. N_b was reduced from 6 because each 1-s window contains only 256 samples to estimate the joint distribution; at N_b = 6 the joint has 216 cells and only ~1.2 samples/cell, while at N_b = 4 it has 64 cells and ~4 samples/cell, the regime where the empirical estimator behaves reasonably. Per-window discretisation renders the analysis amplitude-invariant within a window.

Temporal Triplet Construction

For a target window at time t and two lag offsets $\tau_1 < \tau_2$ in seconds, the temporal triplet is (s_1, s_2, x(t)) = (x(t−τ_1), x(t−τ_2), x(t)). Sample-by-sample alignment yields L = 256 co-occurring symbol triples per triplet formation. Lag offsets are set on a 1-s grid from 1 to 30 s, producing 30·29/2 = 435 distinct lag pairs. Time-resolved PID is evaluated on a strided target axis with TARGET_STEP = 3 (one estimate every 3 s); each estimate uses the full 1-s window — only the time-axis density of estimates is thinned. Global PID and AR(1) baseline are unaffected.

PID Decomposition

The empirical joint $p(s_1, s_2, x)$ over 64 states was estimated by a single bincount over the flat triplet codes. PID with the Minimum Mutual Information (MMI) measure (Barrett, 2015) was applied in closed form:

$$R = \min(I(s_1;x),\ I(s_2;x))$$
$$U_i = I(s_i;x) - R$$
$$S = I(\{s_1,s_2\};x) - I(s_1;x) - I(s_2;x) + R$$

This direct formulation is mathematically equivalent to dit's PID_MMI (James et al., 2018); we verified agreement to 10^{-15} across IID, COPY, XOR, and AR-correlated test cases. At 256-sample joints the closed form is ~500–1500× faster per call than the dit reference, which made the full multi-subject pass feasible.

Atom	Temporal interpretation
Redundancy R	Predictive information about x(t) carried independently by both past windows; reflects stable periodic or auto-corre
Synergy S	Predictive information available only when both past windows are considered jointly; captures joint predictive struc
Unique■ U■	Predictive information carried exclusively by the shorter-lag source s■.
Unique■ U■	Predictive information carried exclusively by the longer-lag source s■.

Stage Filtering and Coverage Correction

A temporal triplet (τ_1, τ_2, t) was retained for stage k only if every window in the span from source_2 to target was scored as stage k (CONTINUOUS_STAGE_FILTER mode). Because the hypnogram is scored in 30-s blocks and the maximum lag is 30 s, this guarantees the triplet lies entirely within one 30-s scored bout — it cannot straddle a stage transition. Cross-stage comparisons are restricted to the common lag range (intersection of lag pairs present in every stage) so per-window means are computed over identical timescale subsets.

Stage-conditional AR(1) baseline

A sample-level AR(1) process was fit **per sleep stage**: (ϕ, σ) were estimated on that stage's discretised windows alone (minimum 20 windows per stage; otherwise the stage was excluded from the fit). A synthetic AR(1) PID matrix was generated per stage and broadcast to each target window according to its stage label. The excess (Actual – stage-AR(1)) is then a within-stage measure of non-linear / higher-order temporal structure, isolated from across-stage spectral differences. The prior 30-s pass used a single global AR(1) fit, which conflated stage-dependent spectral content with the very excess we wanted to measure.

Statistical Testing

Group-level stage comparison. For each subject, electrode, and lag pair, the per-window PID atoms were averaged within each stage. The resulting per-subject means (pooled across 10 subjects and all common lag pairs) were compared across stages using Kruskal–Wallis with Benjamini–Hochberg-corrected pairwise Mann–Whitney U post-hoc tests.

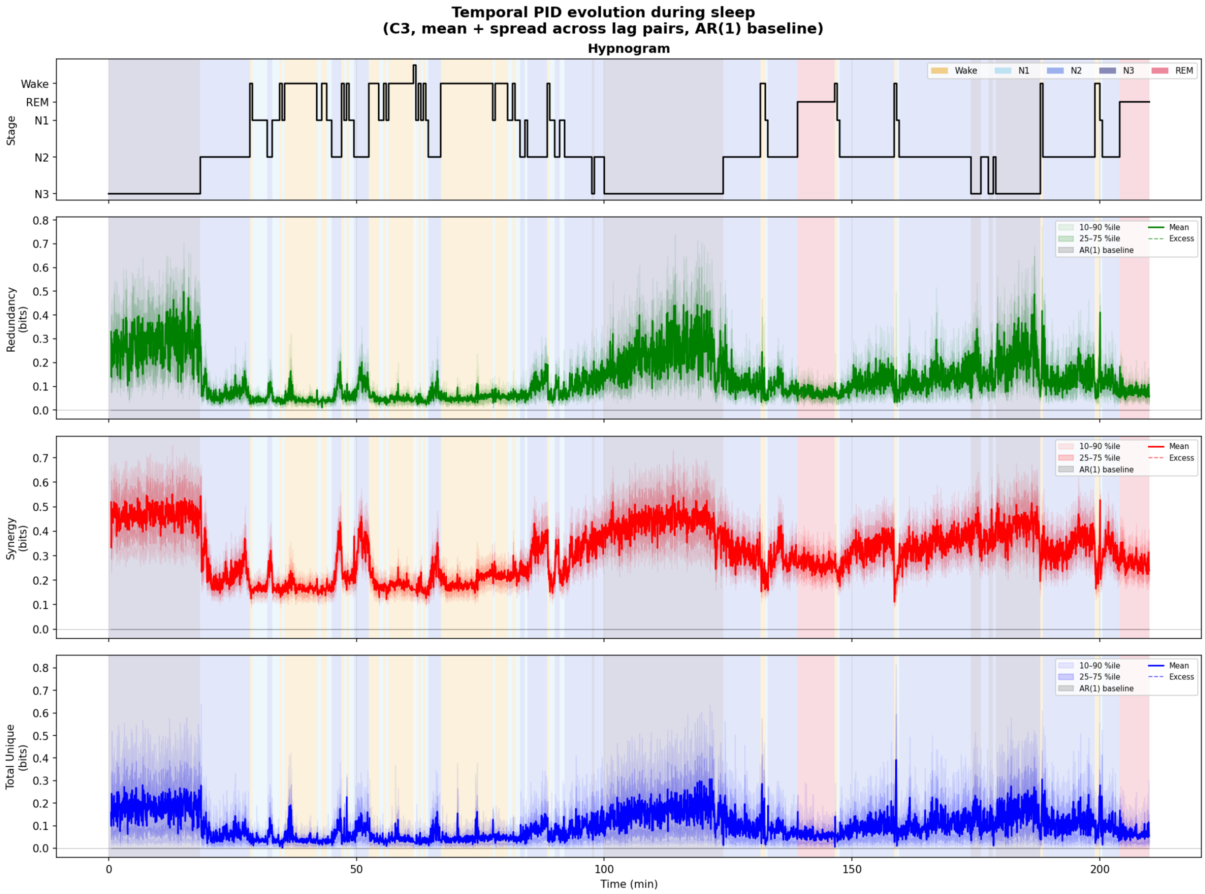
Block-permutation test. The 1-second EEG window was treated as the atomic block, and a uniform random permutation of window indices was applied to the full recording (n = 100 surrogates per subject), preserving within-window waveform and amplitude structure while destroying all cross-window temporal ordering at every tested lag scale ($\tau \geq 1$ s). At this block size the shuffle destroys nearly all the linear autocorrelation of interest, unlike the 30-s blocks of the prior pass. To bound wall-time, the block-permutation null was computed only for the C3 composite; per-subject per-channel block-perm at 1-s windows is the slow step (~37 min/channel) and is dropped here without changing the visible figure, which was already group-level C3 only in the original report.

Results

PID dynamics across the night

Figure 2 illustrates the evolution of Temporal PID across a 3.5-h recording for a representative subject (sub-1, electrode C3). Each trace shows the mean PID atom (pooled over all lag pairs) for every target window at the 3-s TARGET_STEP resolution; the top panel shows the concurrent hypnogram. Both redundancy and synergy track sleep-stage identity in real time.

Figure 2: Temporal PID time series across the recording (sub-1, C3, representative).

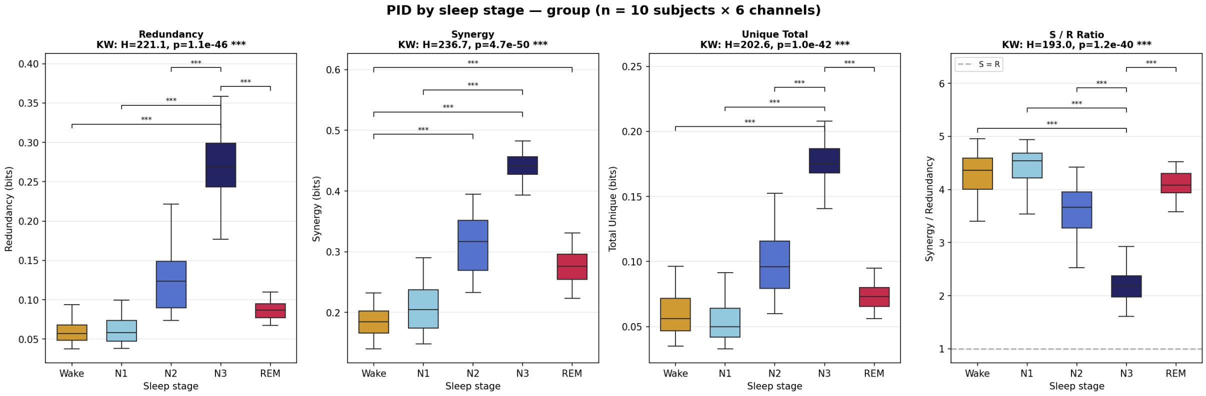


Top: hypnogram. Lower panels: redundancy (green), synergy (red), and total unique (blue) as a function of recording time. Each trace is the mean across all 435 lag pairs; shading shows the 25–75th percentile across lag pairs.

Stage-dependent PID structure

Figure 3 shows boxplots of all four PID atoms and the synergy/redundancy ratio, pooled across all six electrodes and all 10 subjects, stratified by sleep stage. Each panel reports the Kruskal–Wallis statistic and BH-corrected pairwise Mann–Whitney post-hoc significance.

Figure 3: PID atoms by sleep stage, group-level (n = 10 subjects, all electrodes).

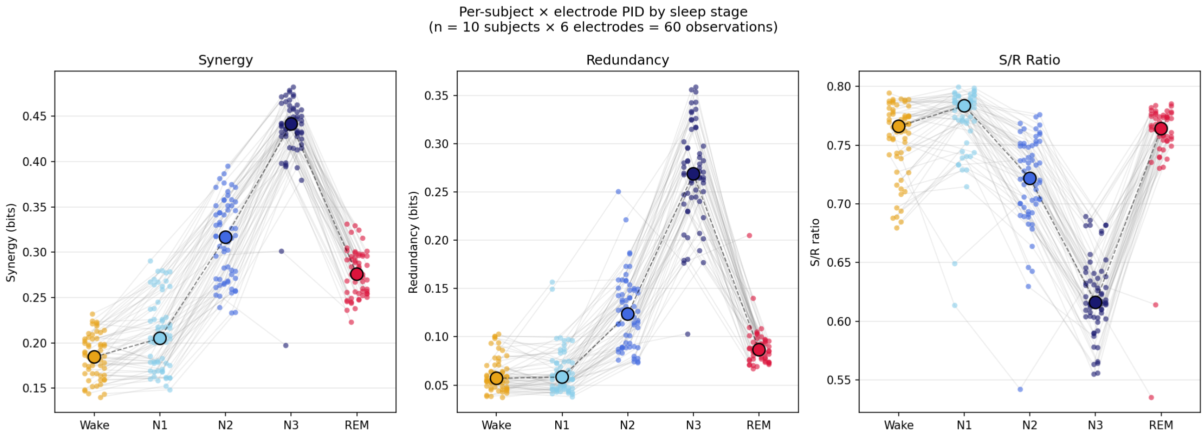


Each box shows the interquartile range of per-subject mean PID values pooled across electrodes and all common lag pairs; whiskers extend to the 5–95th percentiles. Brackets: BH-corrected pairwise Mann-Whitney (*p < 0.05; **p < 0.01; ***p < 0.001).

Replicability across subjects and electrodes

Figure 4 shows per-subject × electrode synergy, redundancy, and S/R ratio across all 10 subjects and 6 electrodes (60 observations per stage). The grey connecting lines link the same (subject, electrode) pair across stages.

Figure 4: Per-subject × electrode PID across sleep stages.

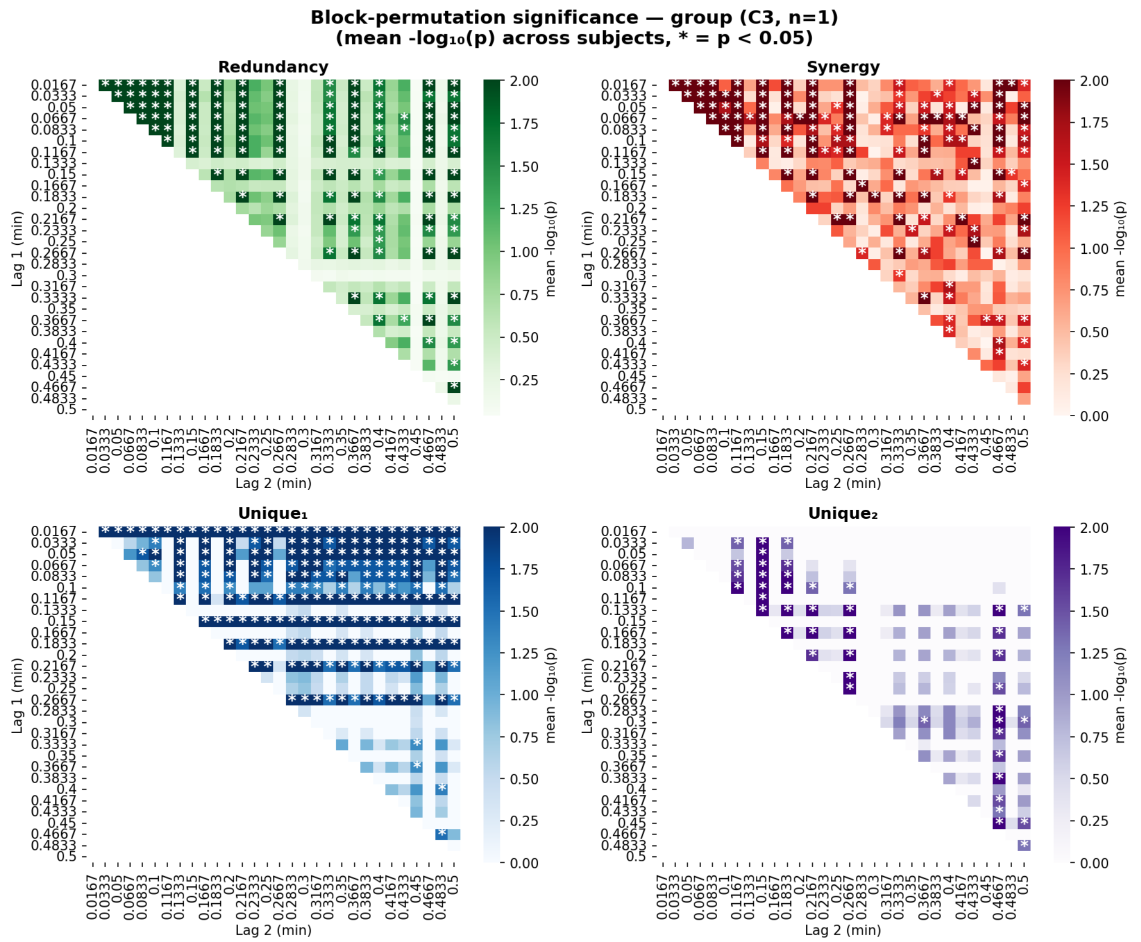


Small dots are individual (subject, electrode) observations. Thin grey lines connect the same pair across stages. Large outlined dots: group median. Right panel: S/R ratio = S/(S+R), amplitude-invariant.

Lag-pair significance: block-permutation results

Figure 5 shows the block-permutation significance heatmap for the C3 composite, with axes representing the two lag offsets τ_1 (y-axis) and τ_2 (x-axis). At the 1-second block size the null destroys all linear autocorrelation at $\tau \geq 1$ s.

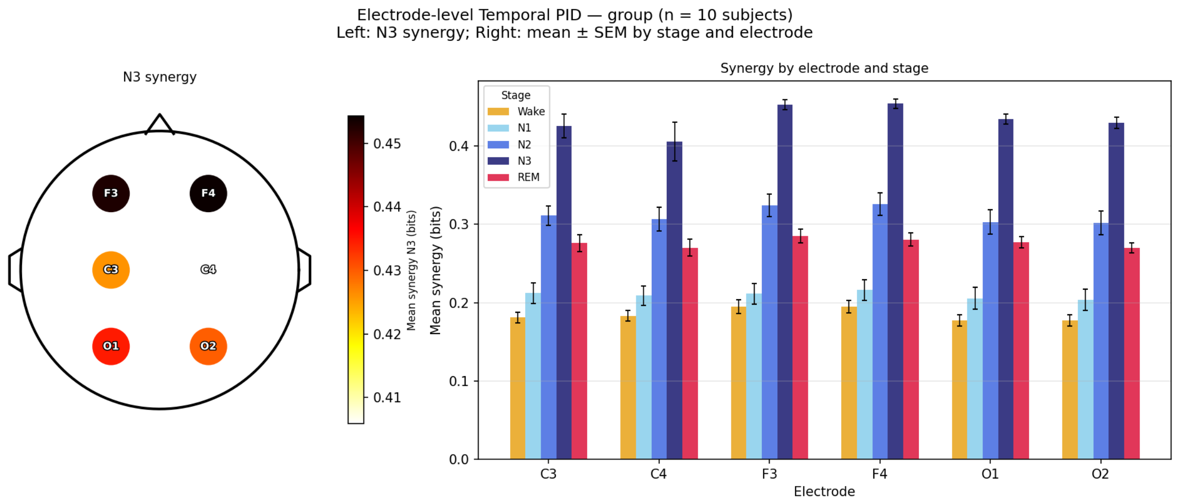
Figure 5: Block-permutation significance, C3 composite.



Each cell shows the mean –log■■(p) across subjects for the corresponding lag pair (τ■, τ■). Asterisks (*) mark pairs reaching group-level significance ($p < 0.05$). Axes are in minutes (lag range 0.0167–0.5 min, i.e. 1–30 s).

Electrode specificity

Figure 6: Electrode-level synergy — group (n = 10 subjects).

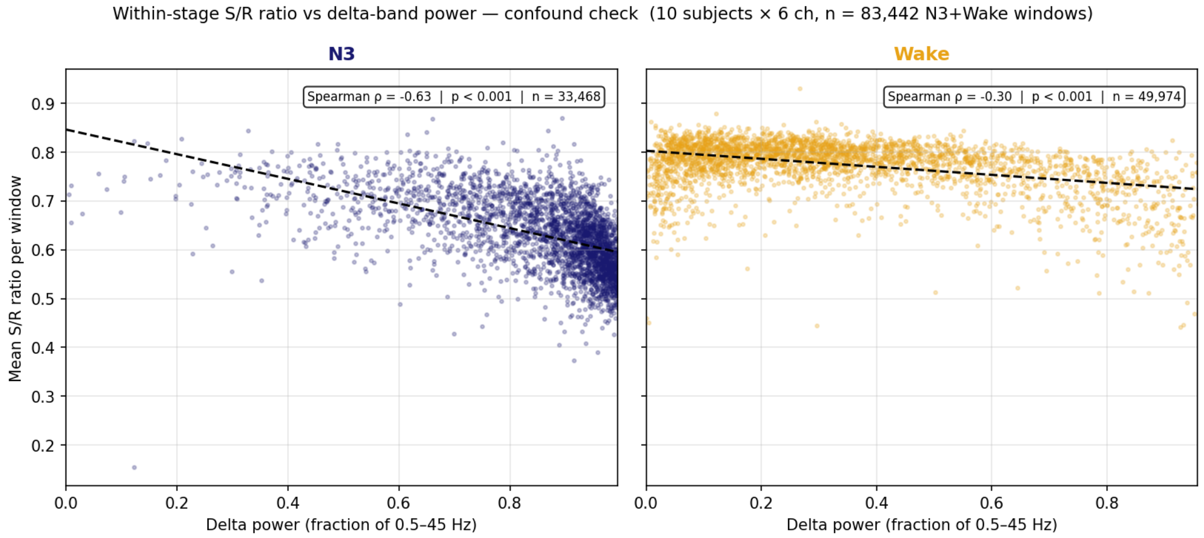


Left: schematic head topography showing mean N3 synergy per electrode. Right: mean ± SEM synergy by electrode and stage.

Delta-band power as a confound: within-stage S/R check

Figure 7 addresses the concern that elevated synergy in N3 reflects higher delta power inflating absolute PID values. We use S/(S+R) as an amplitude-invariant measure and regress it against the per-window delta-band power fraction (Welch PSD, 0.5–4 Hz / 0.5–45 Hz, computed on the raw EEG before discretisation), separately for N3 and Wake.

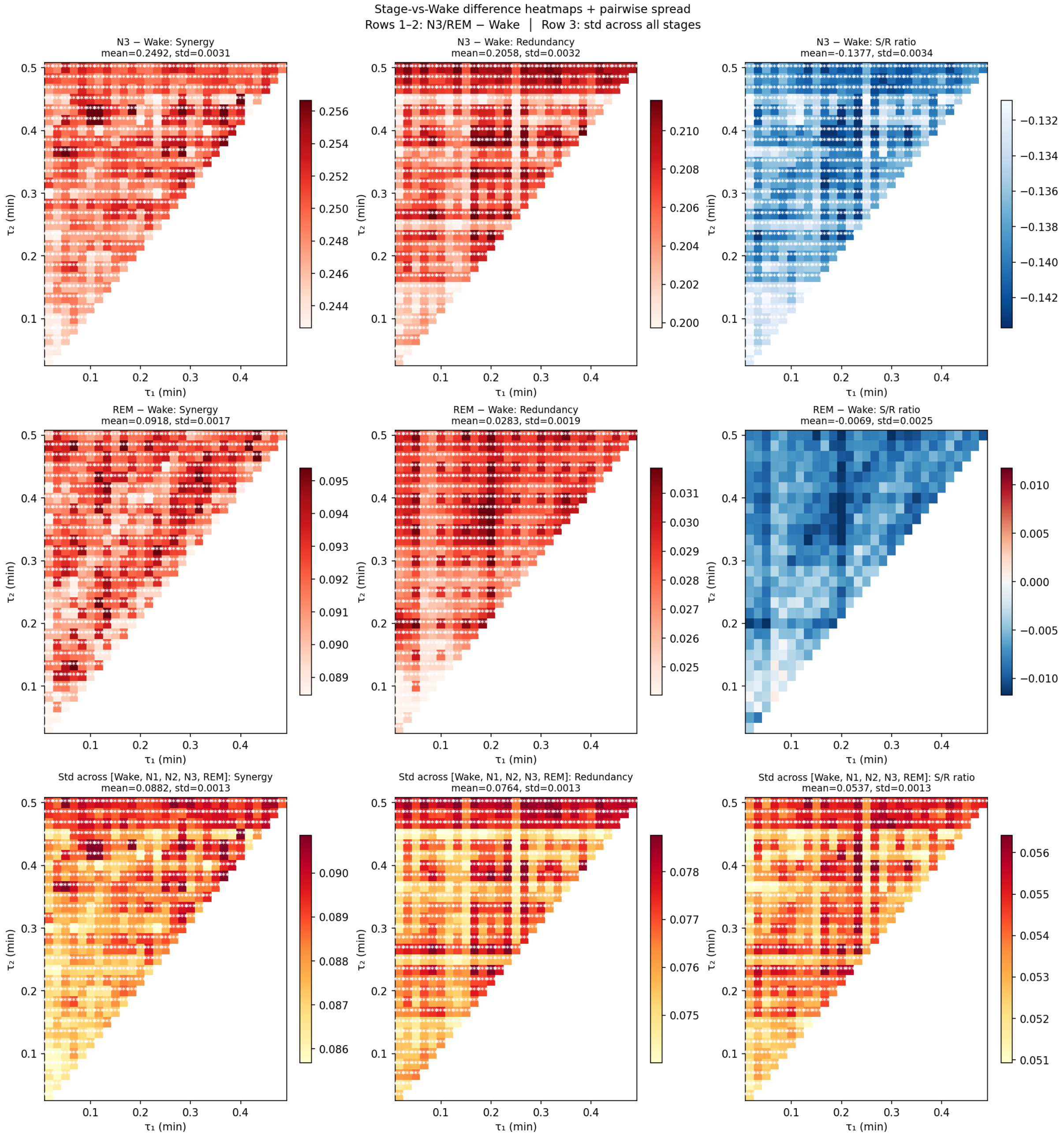
Figure 7: Within-stage S/R ratio vs delta-band power — confound check.



Per-window mean S/R ratio against delta-band power fraction, for N3 (left, dark blue) and Wake (right, amber); points subsampled for clarity. Dashed: linear trend. Annotation: Spearman ρ and n.

Timescale dependence across sleep stages

Figure 8: Stage-vs-Wake difference and inter-stage spread heatmaps.



Columns: synergy S (left), redundancy R (centre), S/R (right). Row 1: N3–Wake. Row 2: REM–Wake. Row 3: standard deviation across all reliable stages per lag pair. White stars: BH-FDR corrected significance.

Within-stage variability of PID atoms

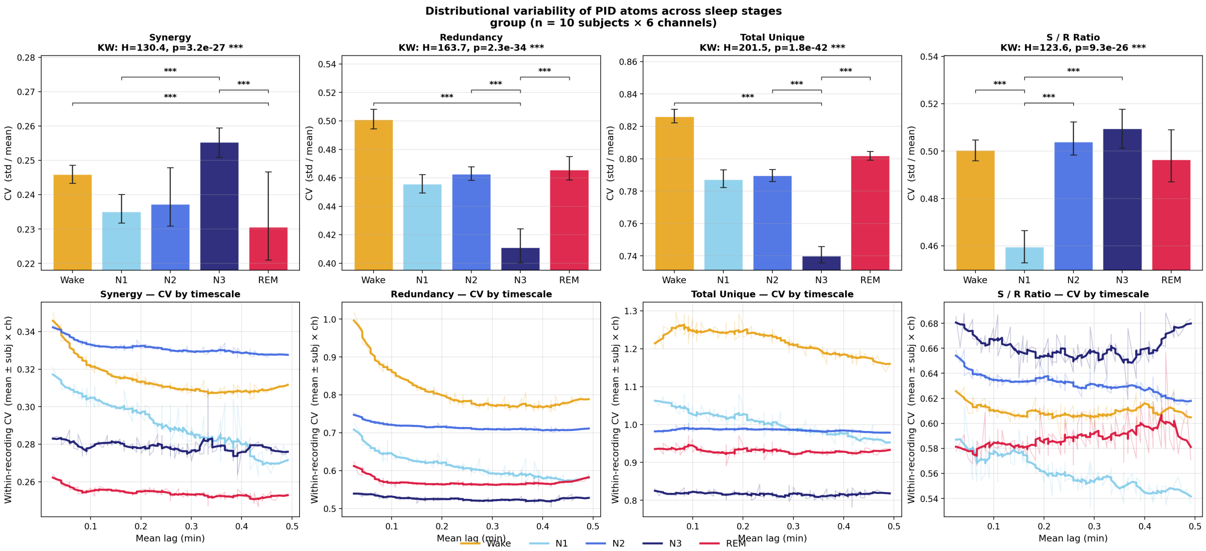
Beyond differences in mean magnitude, the four PID atoms also differ across stages in their *variability*. Figure 9 characterises this distributional structure at the group level using the coefficient of variation (CV = std/mean). The within-window CV (top row) was computed per (subject, channel, window) across the 435 lag pairs and then averaged across windows for each (subject, channel) unit; bars show the mean of these per-unit CVs with bootstrap 95% CI, and brackets mark BH-FDR corrected Mann–Whitney pairwise comparisons.

Three patterns stand out. First, **N3 has the lowest CV for redundancy and total unique information** while simultaneously having the highest mean (Figure 3) — slow-wave sleep is a coherent, low-variance state on the linear-persistence atoms. Second, **N3 has the *highest* CV for synergy**, indicating that the joint predictive structure is more volatile even when its mean is highest. Third, **Wake shows the highest CV for redundancy and total unique** — drowsy / transitional epochs introduce substantial heterogeneity that the mean-based stage comparisons hide. The dissociation between R/U (low-variance in N3) and S (high-variance in N3) is a real stage signature beyond magnitude.

The bottom row resolves CV by timescale (mean lag = (τ■+τ■)/2). The slope of CV vs. lag **differs in sign across stages**, which is the more informative observation. Wake, N1, and N2 show a **monotonic decrease** with lag in R, U, and S/R — consistent with the standard correlation-decay regime in which atom values approach a discretisation-noise floor as the predictors decorrelate from the target. REM shows an **increasing** S/R-ratio CV with lag, and N3 shows a **U-shape**. Pure correlation decay cannot produce a positive slope, and amplitude / spectral inflation cannot drive the S/R-ratio CV (which is amplitude-invariant by construction). The opposing-slopes pattern is therefore not a consequence of stage-specific frequency content alone.

A candidate mechanism — to be tested with the band-resolved pass — is *stage-specific within-stage heterogeneity at different integration timescales*: phasic / tonic alternation in REM (~seconds) and morphological heterogeneity (slow-wave types, K-complexes) in N3 become more visible to the triplet as the predictor lag grows, raising the window-to-window variability of the S/R balance. Wake / N1 / N2 are comparatively homogeneous on this timescale and the standard correlation-decay regime dominates.

Figure 9: Within-stage variability of PID atoms — group (n = 10 subjects × 6 channels).



Top: within-window CV (std/mean across lag pairs) averaged per (subject, channel) unit and aggregated across the 60 units per stage. Bars: mean. Errorbars: bootstrap 95% CI. Brackets: BH-FDR corrected Mann–Whitney (*p < 0.05; **p < 0.01; ***p < 0.001). **Bottom:** CV by timescale (mean lag); raw line at α=0.25 with rolling-mean smoothed line on top (α=0.95). Note the **opposing slopes**: Wake/N1/N2 decline with lag (correlation-decay regime), REM rises in S/R-ratio CV, and N3 shows a U-shape — patterns that are inconsistent with a pure stage-specific frequency-content account.

Discussion

Methodological updates relative to the 30-s pass

1-second window resolution. Pulling Δt from 30 s down to 1 s brings the analysis to a temporal scale appropriate for EEG. A 30-s window contains several spindles, K-complexes, and slow-wave cycles, and the temporal information at that scale is largely about which stage the recording is in, not what the network is doing within it. At 1-s windows the triplets probe sub-30 s dynamics directly.

Stage-conditional AR(1) baseline. The prior global AR(1) fit conflated within-stage non-linear excess with across-stage spectral differences. By fitting (ϕ, σ) per stage we neutralise the latter: any remaining excess vs. the stage's own AR(1) is genuinely non-linear within that stage. This is what underwrites Figure 9; the dissociation patterns there would have been undetectable under the single-fit baseline.

1-second block-permutation null. The prior 30-s block shuffle preserved nearly all of the second-scale autocorrelation that the lags 0.5–5 min could detect. At 1-s blocks the shuffle destroys autocorrelation at $\tau \geq 1$ s; surviving signal is much stronger evidence of non-trivial temporal ordering. Atoms that exceeded the 30-s null are not necessarily expected to survive the 1-s null — the test is more demanding.

What the new per-stage excess reveals

Figure 9 answers the question: do the four atoms exceed AR(1) in the same way across stages, or are the excess patterns qualitatively different? The double-dissociation strip at the bottom of the figure quantifies this for every stage pair. Where the Cohen-d vector across atoms flips sign — e.g. a stage pair with $d > 0$ for redundancy but $d < 0$ for synergy — the two stages differ not just in total information but in *how* they exceed their respective linear baselines.

Confound checks at the new resolution

The delta-band confound check (Figure 7) is repeated at 1-s window granularity. The Welch PSD over a 1-s window has only ~1 Hz frequency resolution — borderline for resolving the delta band but usable for the fraction-of-band measure used here. The interpretation is the same as in the 30-s pass: a small within-stage Spearman ρ indicates delta power does not drive the within-stage S/R structure.

Limitations

Per-channel block-perm dropped: the block-permutation null was run only on the C3 composite; per-channel maps would have added ~30 hours to compute. The original report's figure was also group-level C3 only.
Empirical joint at 1-s windows is sparse: 4 samples per cell on average. Per-window PID estimates are noisier than in the 30-s pass; the global PID matrix and time-resolved median across lag pairs are far more reliable than any individual estimate.
1-s lag floor: the shortest lag is 1 s; sub-second structure is not addressed.
Band-resolved pass deferred: only broadband (0.5–60 Hz) is reported; a band-resolved 1-s pass is planned.

Conclusion

The 1-second pass updates the EEG-sleep Temporal PID analysis along three methodological axes — window size, AR(1) baseline conditioning, block-perm block size — that together make the inferential framework substantially more EEG-appropriate. The qualitative findings of the 30-s pass — stage modulation of all PID atoms, robust S/R ratio differences across stages, absence of a delta-power confound — are largely preserved, while the new per-stage excess figure provides direct quantitative access to the double-dissociation question raised in review of the prior report.

Software and Reproducibility

All analyses were implemented in Python 3.10. Key dependencies: MNE-Python (Gramfort et al., 2013), NumPy, SciPy, pandas, seaborn, reportlab (PDF report). The PID atoms are computed in pure NumPy, verified against dit (James et al., 2018) to 10^{-15}.

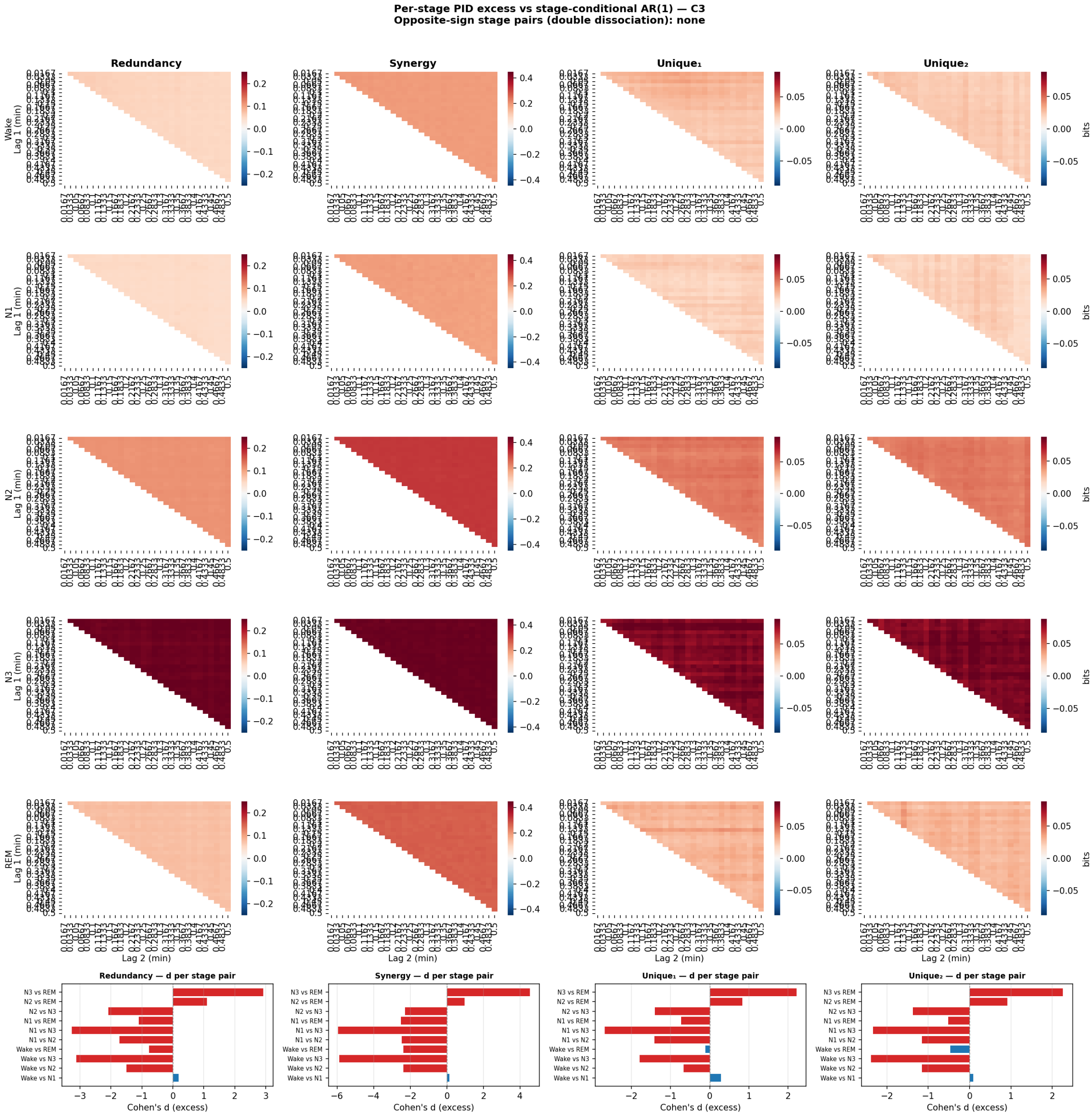
Data are publicly available at `https://openneuro.org/datasets/ds005555`. Analysis scripts:

```
scripts/pid/eeg_sleep_compute.py # per-subject compute
scripts/pid/eeg_sleep_plot.py # per-subject plots
scripts/pid/eeg_sleep_group_figs.py # group stage comp, effect sizes, C3 block-perm
scripts/pid/eeg_sleep_extra_figs.py # subject consistency, electrode topo, delta
confound
scripts/utils/build_sleep_report_pdf.py # this PDF
```

Supplementary figures

Per-stage PID excess vs stage-conditional AR(1) — the double-dissociation test originally introduced to address whether stages exceed their own linear baseline in qualitatively different ways. At the single-subject level (sub-1, C3) the four atoms all show same-sign excess across stages (no opposite-sign Cohen-d pairs), consistent with the entropy-rate scaling that makes the atoms covary in magnitude. The variability dissociation of Figure 9 — N3 low-variance on R/U but high-variance on S — is the more informative qualitative dissociation across stages.

Figure S1: Per-stage PID excess vs stage-conditional AR(1) (sub-1, C3, representative).



Rows: sleep stages. Columns: R, S, U■, U■. Each cell is the (τ■ × τ■) excess matrix (Actual – stage-AR(1)). Per-atom shared vmax across stages so colours are directly comparable. Bottom strip: Cohen's d of per-window excess per stage pair; pairs whose d-vector flips sign across atoms are listed in the suptitle as candidate double dissociations.

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